御杖村 · BIOMASS ENERGY & AI

Forest → Power → Compute: the material and money loop

One map of the whole system — how wood, heat and labour move through it (green track), and where money is invested and returned (money track) — for the doubled-crew baseline (~1.15 MWe).

Planning-grade / illustrative · 2026-07-17 · figures from the project's own docs, to be firmed in Phase 1 feasibility. No funding committed yet.

- Materials & labour
- Electricity
- Heat
- Compute
- Money in (investment)
- Money out (revenue & returns)

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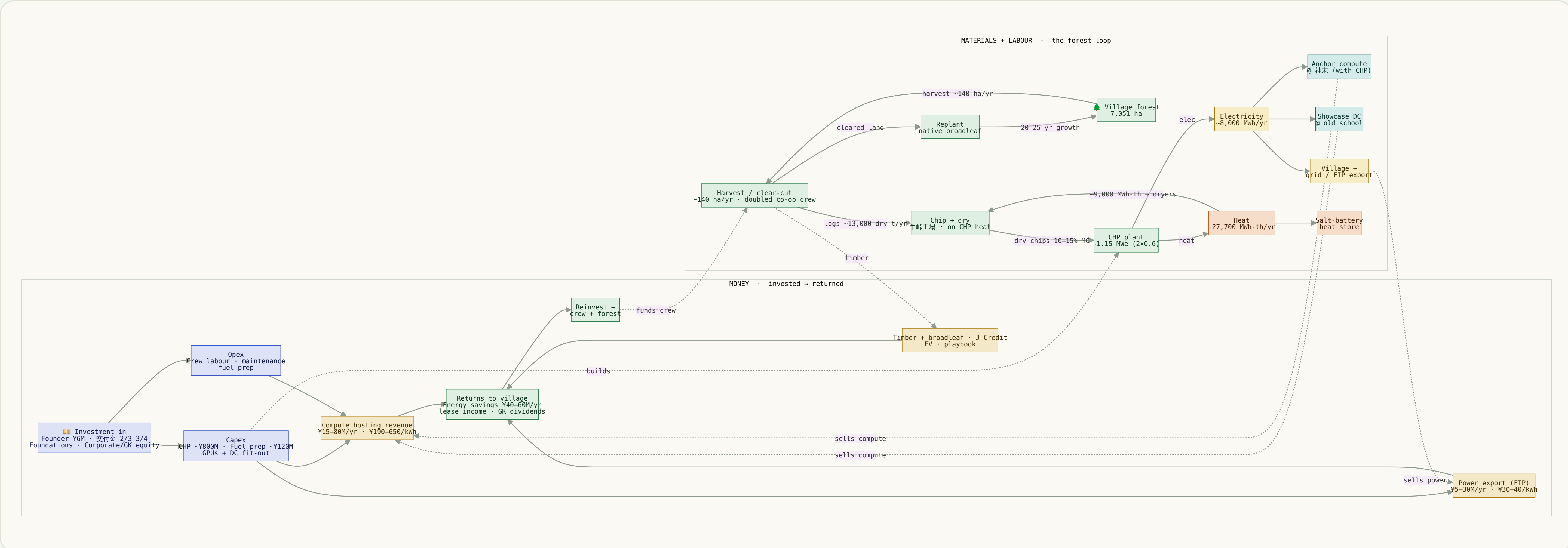
01 The whole loop, one graph

Solid arrows are physical flows and money flows; dotted arrows couple the two — an asset to the money that builds it, a use to the revenue it earns, and the reinvestment that closes the loop back into the forest.



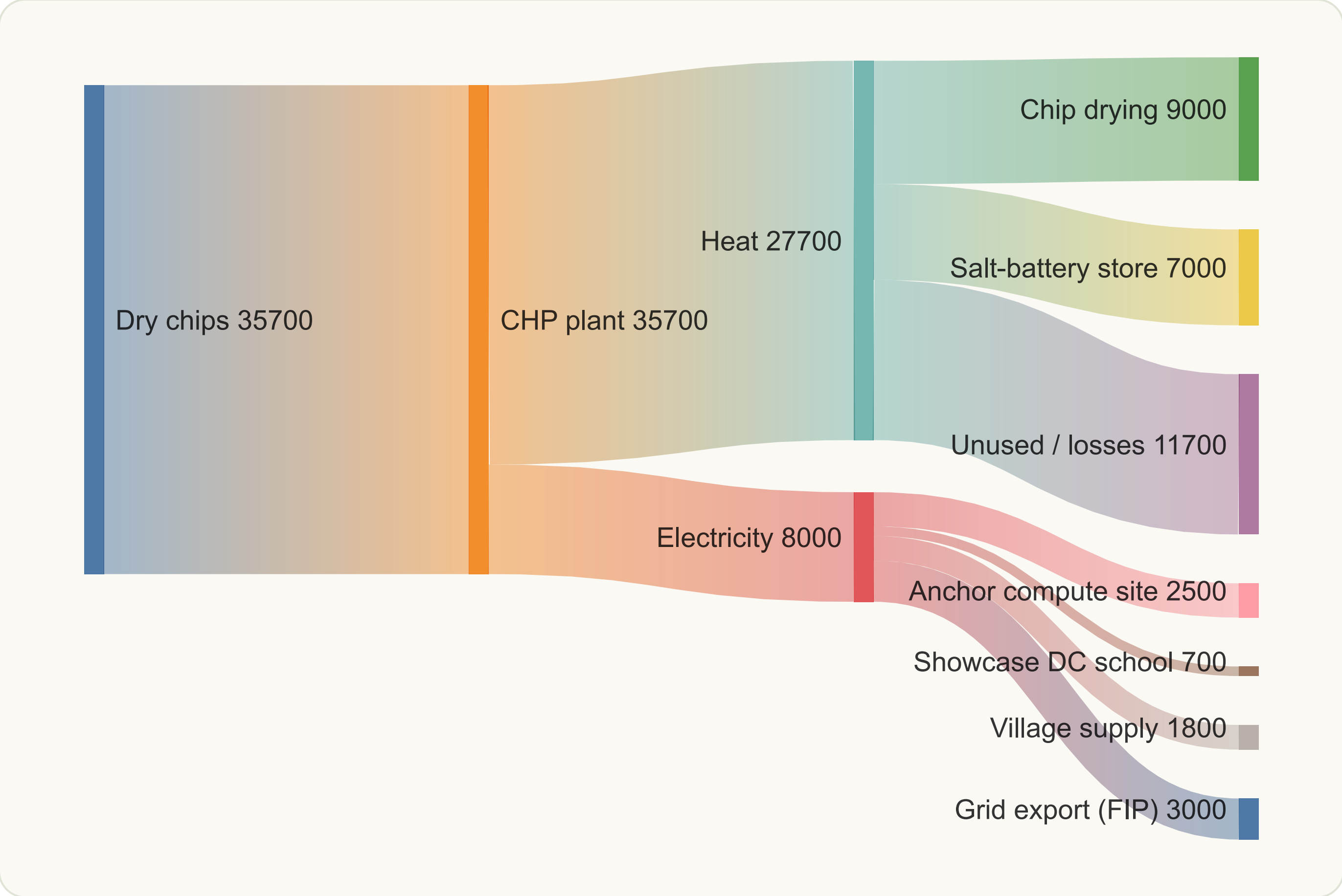
Static render for print/PDF — the interactive zoom/pan version lives in the project's Claude artifact.

The one number that drives the shape: a kWh is worth **~6–20× more as AI compute (¥190–650) than as exported power (¥30–40 FIP)**. So the CHP's best customer is an on-site GPU data centre, not the grid — compute is the offtake, FIP export is the overflow valve, and reinvested surplus grows the crew that grows the harvest.

02 Same flows, widths to scale (Sankey)

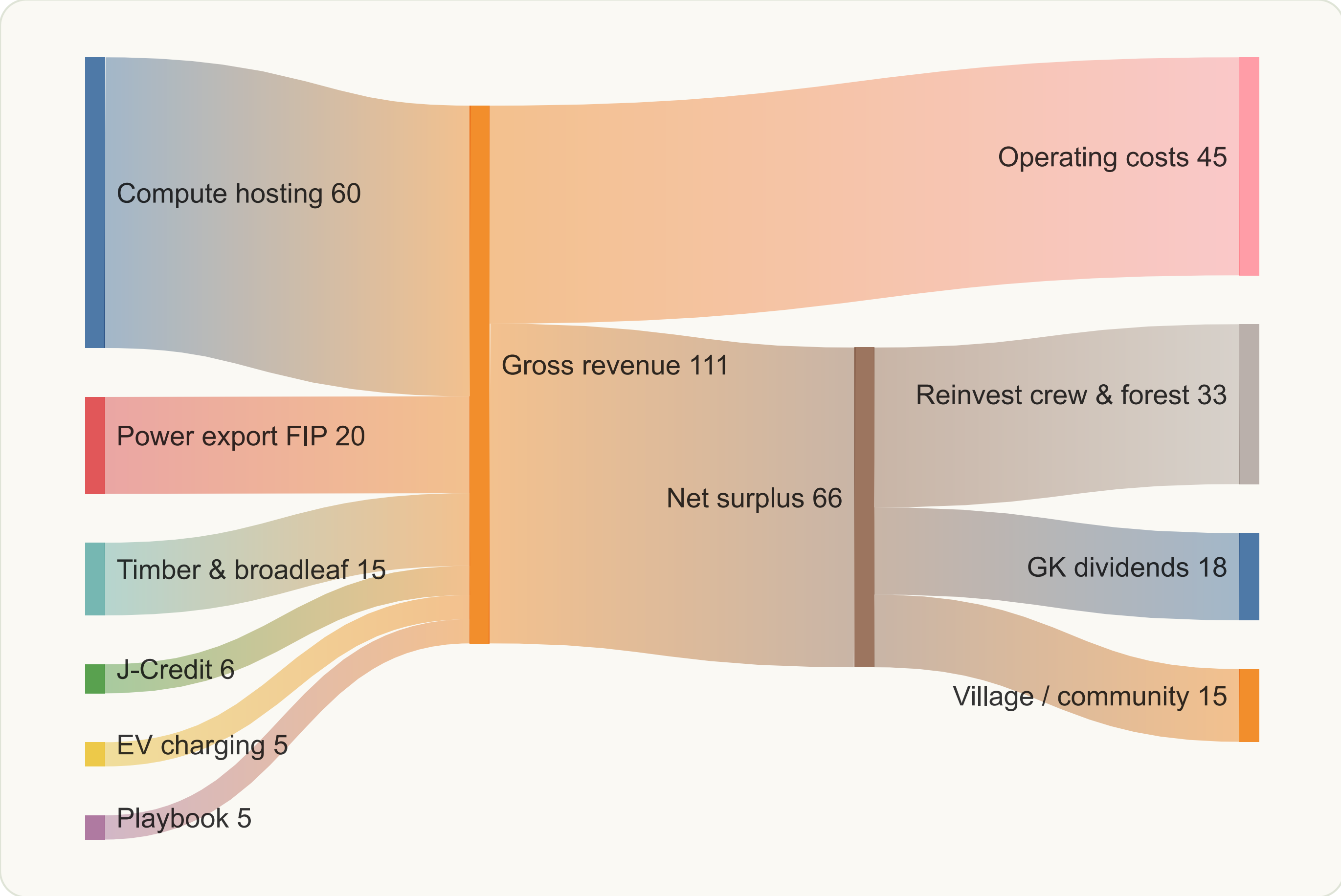
Two Sankey diagrams where band thickness = magnitude. Left: energy in MWh/yr. Right: money in ¥M/yr (Year-10 midpoints). The electricity split is an **illustrative** allocation — the compute share is the design lever we're still sizing.

Energy · MWh/yr



Chips→CHP→elec+heat is measured; the four electricity uses are an illustrative split of the ~8,000 MWh.

Money · ¥M/yr (Yr 10)



Year-10 midpoints; opex ≈ 40% of revenue; surplus split illustrative.

03 The numbers behind the arrows

Authoritative figures for the two tracks. Material balance is the doubled-crew baseline; money is illustrative Year-5→Year-10 framework, not a forecast.

Material & energy balance

Money in · Money out

Doubled-crew baseline · annual · from the forest-workforce plan

STAGE	QUANTITY
Village forest (total)	7,051 ha
Harvest / clear-cut rate	~140 ha/yr
Forest worked over 25 yr	~3,526 ha (50%)
Fuel to plant	~13,000 dry t/yr
CHP capacity	~1.15 MWe
Electricity	~8,000 MWh/yr
Heat (total)	~27,700 MWh-th/yr
— of which to drying	~9,000 MWh-th/yr
Replant	native broadleaf

Investment (one-off) vs annual revenue · illustrative

IN — CAPITAL	¥
Programme target (Ph 0–3, BAC ¥220M)	~192M
CHP gensets 2×0.6 MWe (Ph 4)	~800M
Fuel-prep line (Ph 4)	~75–215M
GPUs + DC fit-out	demand-led
OUT — REVENUE (YR 5 → YR 10)	**¥/YR**
Compute hosting	15–30M → 40–80M
Power export (FIP)	5–15M → 10–30M
Timber / broadleaf	3–8M → 10–20M
J-Credit · EV · playbook	3–9M → 9–26M
Total revenue	**28–67M → 74–166M**
+ Village energy savings	40–60M/yr

Sources: `docs/energy-forest/mitsue_forest_workforce_energy_plan.md` (material/energy, capex), `docs/strategy/mitsue_revenue_model.md` (revenue & displacement), `docs/energy-forest/mitsue_fit_grid_check.md` (FIP vs compute, siting), `docs/strategy/mitsue_evm_plan.md` (BAC ¥220M). GPU pricing: GPUSOROBAN / HIGHRESO. All figures are the project's own planning-grade estimates — Phase 1 feasibility (Dec 2026) replaces them with vetted numbers. Nothing is committed funding.